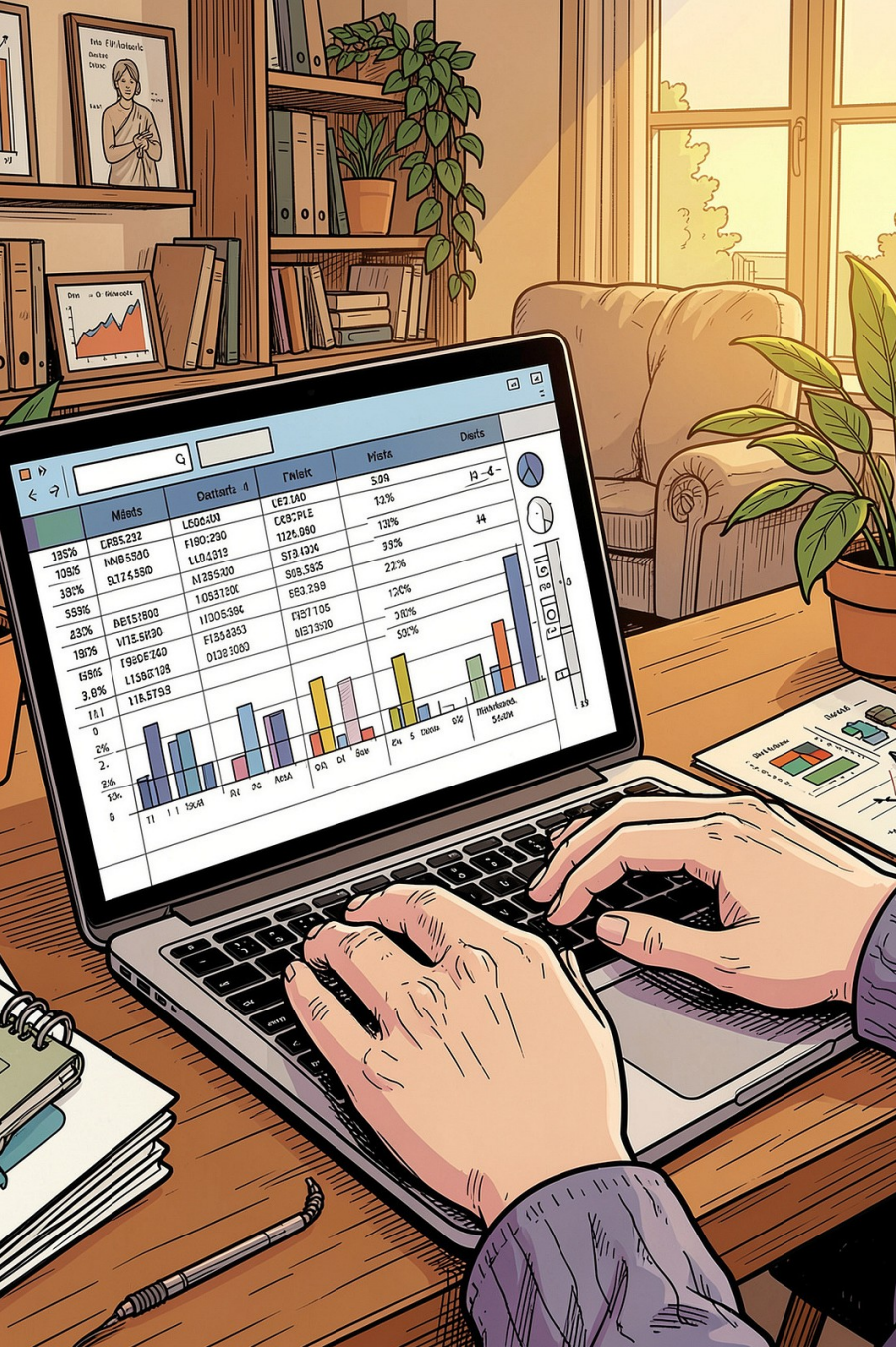




Customer Shopping Behavior Analysis

3,900 purchases • 18 features • actionable insights



Project Overview

Goal

Uncover spending patterns, segments, product & subscription behavior

Data

3,900 rows · 18 columns · multi-category transactions

Outcome

Recommendations for subscriptions, loyalty, pricing, targeting

Dataset Snapshot



The illustration shows a hand-drawn table with several columns and rows. The columns are labeled with various attributes, some of which are repeated or misspelled. The data points are also hand-drawn and somewhat inconsistent.

CUSTOMER ID	AGE	AGE	ITEM	AMOUNT	AMOUNT	AMOUNT
GENDER						
GENDER						
AMOUNT						
AMOUNT						
AMOUNT						

Additional labels and data points visible in the illustration include:

- Labels: CUSTOMER ID, AGE, AGE, ITEM, AMOUNT, AMOUNT, AMOUNT, GENDER, GENDER, AMOUNT, AMOUNT.
- Data points: 15X, 170, 25, 39, 39, 8, 20.

Key features

- Demographics: Age, Gender, Location, Subscription
- Purchase details: Item, Category, Amount, Season, Size, Color
- Behavior: Discount, Promo, Frequency, Reviews, Shipping

Missing: 37 review ratings

Data Preparation (Python)

01

Load & Explore

pandas · df.info() · .describe()

02

Clean & Impute

Median per category → Review Rating
(37 nulls)

03

Engineer

age_group, purchase_frequency_days

04

Standardize

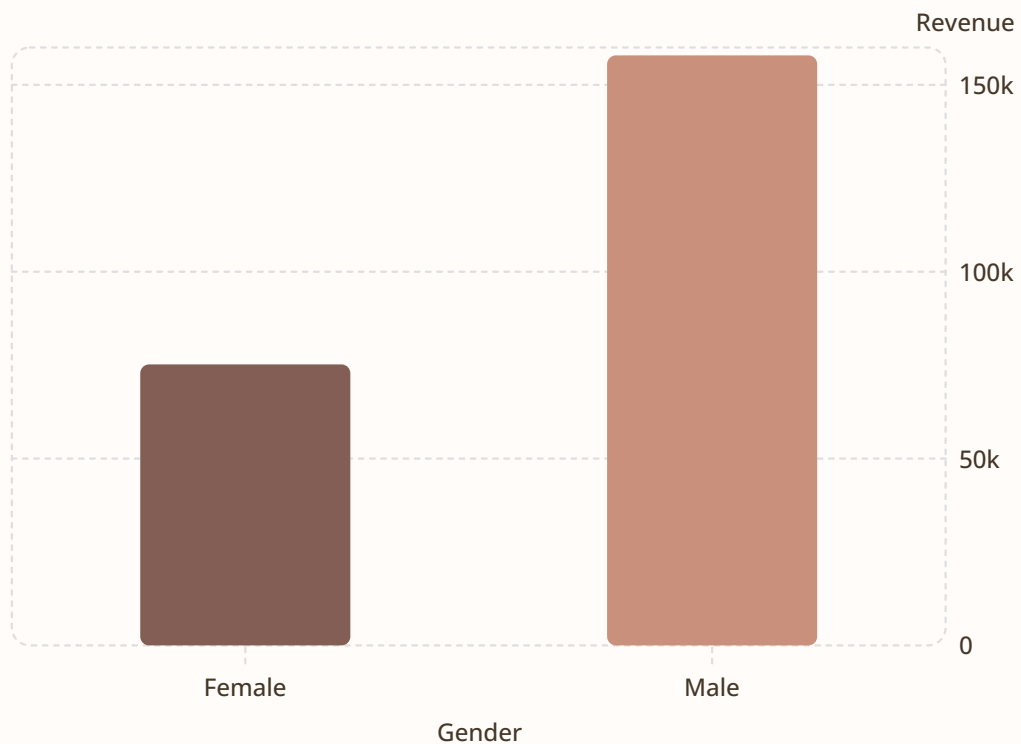
snake_case column names

05

DB Load

PostgreSQL integration for SQL analysis

Revenue & Spend Insights (SQL)



Highlights

- Male revenue > Female (157,890 vs 75,191)
- Avg purchase $\approx$ \$59.76
- Avg review rating 3.75

Top Products & Discount Patterns



Top Rated

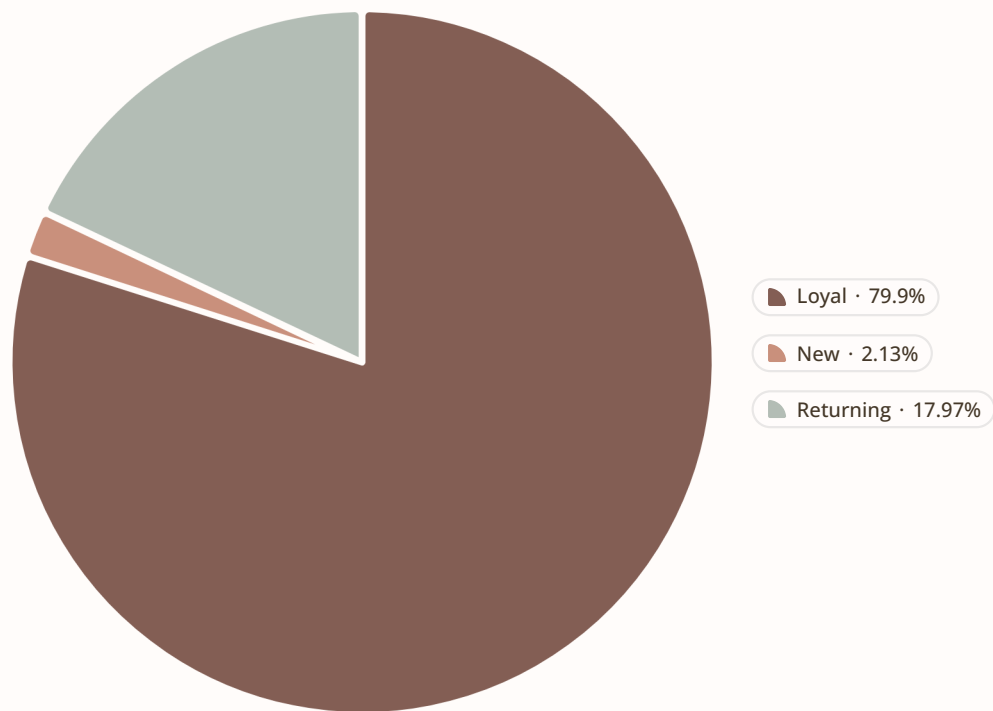
Gloves, Sandals, Boots, Hat, Skirt (top 5 by avg rating)



Discount-Dependent

Hat 50% · Sneakers 49.66% · Coat 49.07% ·
Sweater 48.17% · Pants 47.37%

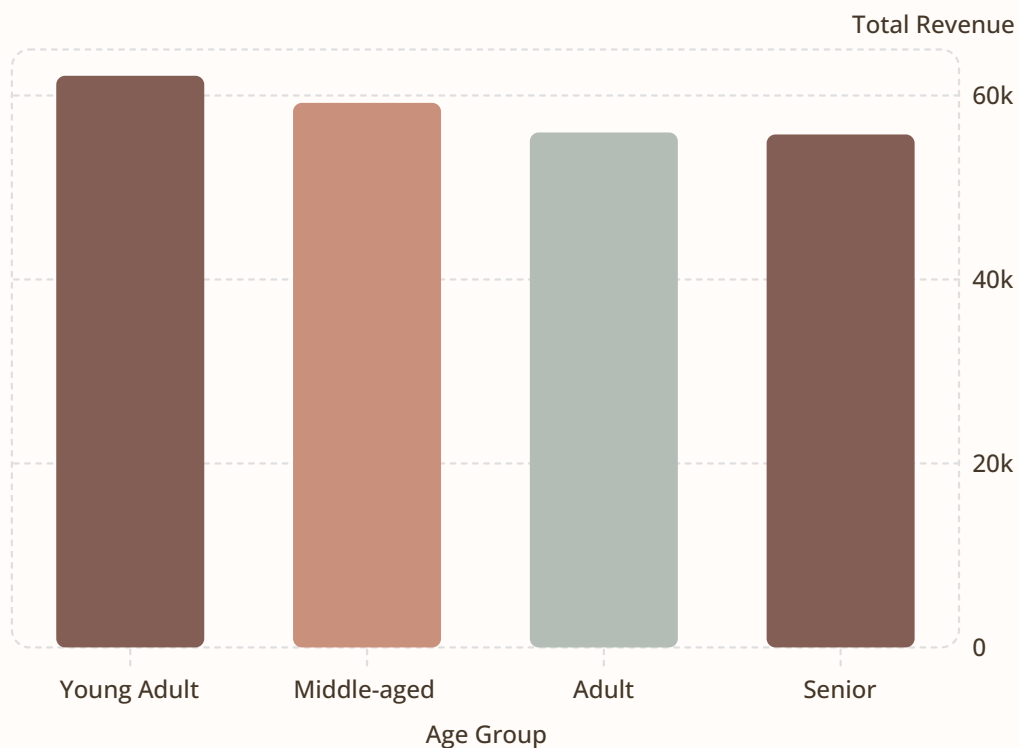
Customer Segments



Notes

- Loyal: 3,116 customers (majority)
- Repeat buyers (>5): more likely to subscribe
- Subscribers: 1,053 · Non: 2,847

Age & Shipping Effects



Shipping

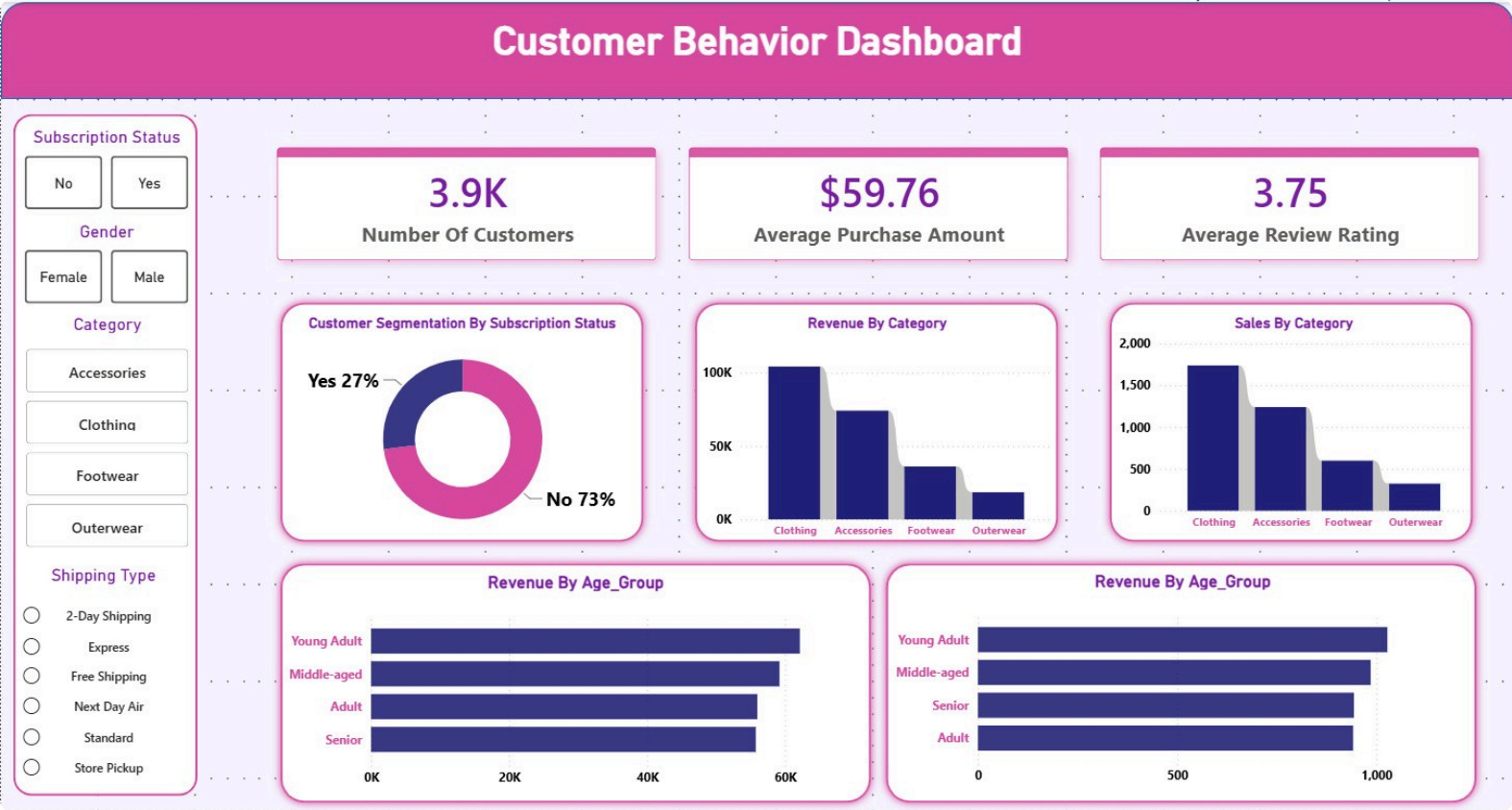
Standard avg \$58.46 · Express avg \$60.48

Targeting

Focus marketing on Young Adults & Middle-aged (top revenue groups)

Dashboard Preview

Interactive Power BI dashboard — key metrics at a glance



3.9K customers · \$59.76 avg purchase · 3.75 avg rating · 27% subscribers

Business Recommendations

Boost Subscriptions

Promote exclusive benefits to raise conversion

Loyalty Programs

Reward repeat buyers to grow Loyal segment

Review Discounts

Balance discounting with margins; target discount-dependent products

Product Positioning

Feature top-rated & best-selling items in campaigns

Targeted Marketing

Prioritize high-revenue age groups and express-shipping users